

Technology Stack (Architecture & Stack)

Phase / Category	Requirement Analysis	Date	1/10/2026
Team ID	6ab9466dc23107aaaa73904b	Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks		

Technology Stack (Architecture & Stack)

This deliverable defines the technical architecture, platform modules, runtime libraries, and architectural characteristics of the solution implemented within the ServiceNow ecosystem.

1. Technical Architecture Overview

The solution operates on a 3-tier enterprise architecture consisting of a Presentation Tier (Client Browser / GlideForm), an Application Server Tier (ServiceNow Jakarta/Washington PaaS), and a Persistence Tier (ServiceNow Multi-tenant Relational Database).

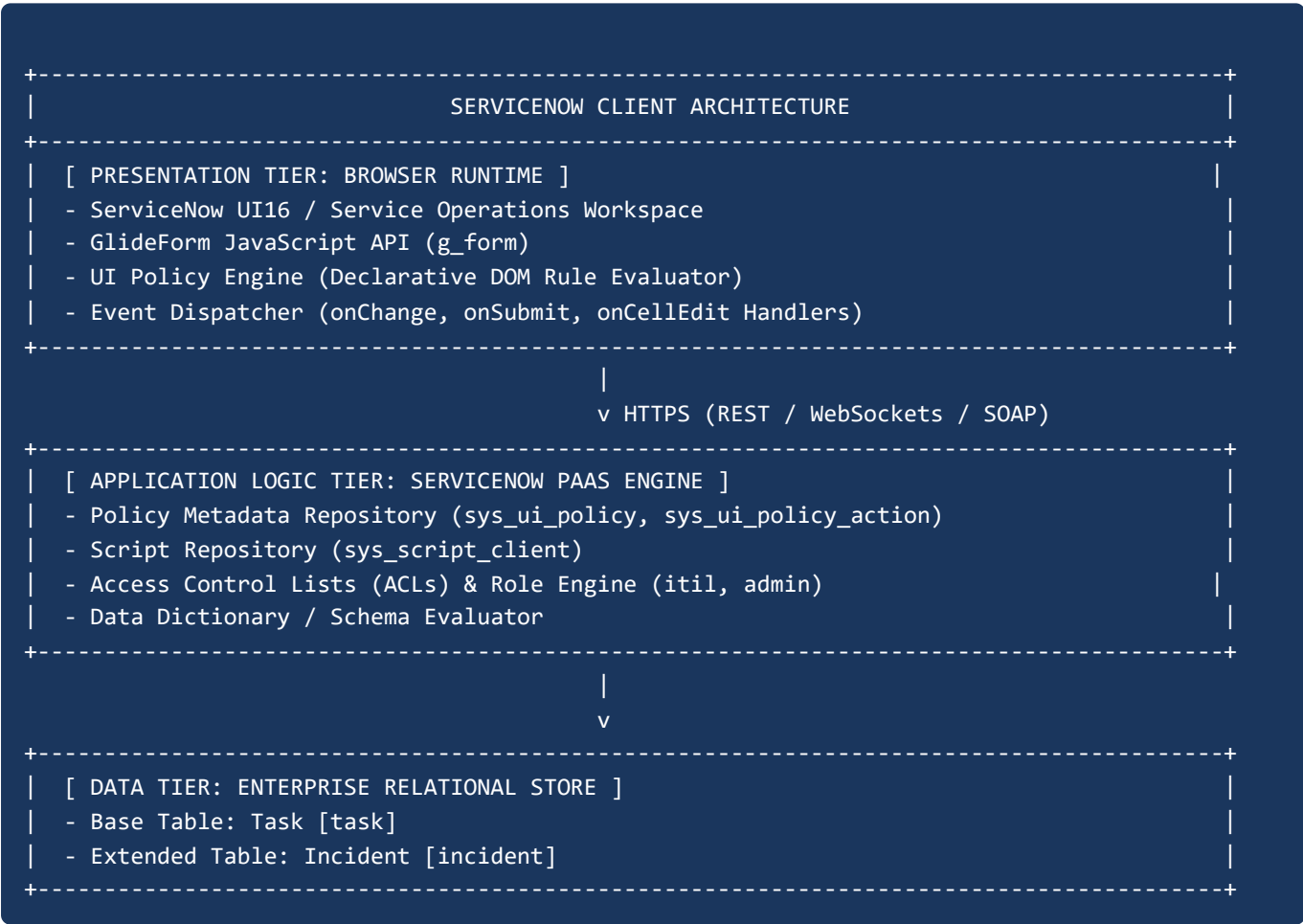


Table-1: Components & Technologies

S.No	Component	Description	Technology / Specification
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1	User Interface	Interactive Incident Form and Incident List View rendered in desktop browser.	HTML5, CSS3, JavaScript (ES6), ServiceNow GlideForm (g_form) client API.
2	Declarative Logic Engine	Rule-based client configuration engine managing dynamic mandatory & read-only field states.	ServiceNow UI Policy Engine (sys_ui_policy, sys_ui_policy_action).
3	Procedural Client Scripts	Event-driven JavaScript functions executing on field modification, form save, and list edit.	ServiceNow Client Scripts (onChange, onSubmit, onCellEdit).
4	Feedback Components	Inline field alerts and top-level informational messages.	g_form.showErrorBox(), g_form.addInfoMessage().
5	Database / Data Model	Core ITSM schema holding incident attributes, priorities, assignments, and audit logs.	ServiceNow Relational Database (MariaDB cluster), Table: incident.
6	Infrastructure / Cloud	High-availability enterprise multi-instance cloud platform.	ServiceNow Enterprise SaaS Cloud (Washington / Xanadu release).

Table-2: Application Characteristics

S.No	Characteristics	Description & Justification	Technology Implemented
1	Security Implementations	Role-based access control restricts configuration access to administrators while allowing ITIL users to benefit from form guardrails. List edit restrictions prevent state bypass.	ServiceNow Access Control Lists (ACLs), Role-Based Access Control (itil, admin), onCellEdit script security.
2	Scalable Architecture	All field validations and automation run directly in the client's browser, consuming zero application server CPU cycles during form typing.	Client-side DOM execution, GlideForm JavaScript API.
3	High Availability	Cloud-hosted across geodistributed data centers with automated failover and 99.99% enterprise uptime.	ServiceNow Advanced High Availability (AHA) multi-instance architecture.
4	Performance	Sub-50ms reaction time for field validations; eliminates unnecessary server roundtrips, ensuring instant responsiveness during high-volume ticket logging.	Client-side event listeners and local DOM evaluation.